	Target engagement	Dose	Efficacy	Safety
Metabolic	**mucopolysaccharidosis type 1** IDUA replacement / gene therapy (4)	**2022** MPS I dogs tolerized to human IDUA, intracisternal… △ No lasting IDUA in patients	**2018** Rhesus monkey PK and chronic toxicology, AGT-181 Dosed to 6 mg/kg within margin	**2023** Idua-/- mouse, ZFN albumin-locus IDUA insertion Not measured in patients	**2021** NSG mice engrafted with human (incl. MPS I patient) HSPCs transduced with clinical-grade IDU…
	mucopolysaccharidosis type 3B NAGLU replacement / gene therapy (3)	**2023** MPS IIIB dog, ICV tralesinidase alfa (20 months) △ CSF HS change small, not…	**2010** MPS IIIB dog, intracerebral rAAV5-hNAGLU with… No checkable prediction	**2023** MPS IIIB dog, ICV tralesinidase alfa (20 months) Patient result unclear	**2017** MPS IIIB dog, intracerebral rAAV5-hNAGLU with immunosuppression
Neurology	**Duchenne muscular dystrophy** DMD (8)	**2011** DMD patient-derived myoblasts/myotubes and human muscle cell lines (sequence…	**2016** Rodent and cynomolgus monkey GLP toxicology Model data unpublished	**2020** DMD patient-derived myoblasts/myotubes and human… ☐ Outcome trial negative	**2020** Rodent and cynomolgus monkey GLP toxicology Again no kidney toxicity in patients
	Huntington disease HTT antisense / siRNA (4)	**2022** HD patient iPSC-derived neurons and astrocytes (HD71, HD180) on AAV5-miHTT	**2019** HD rodents and tgHD minipig (AAV5-miHTT dose response) No checkable prediction	**2025** HD rodents (Q175 knock-in, R6/2) on intrastriatal AAV5-miHTT △ cUHDRS worse on tominersen	**2022** NHP and rat intrastriatal AAV5-miHTT toxicology High-dose inflammatory SUSARs
	fragile X syndrome GRM5 negative modulators (3)	**2009** Fmr1 KO mouse prepulse inhibition (PPI) of startle	**2005** Fmr1 KO mouse behaviour and spine rescue No checkable prediction	**2011** Drosophila dfmr1 mutant (courtship memory, mushroom body) Mavoglurant missed ABC-C primary	**2008** Sponsor GLP toxicology (rodent and non-rodent) Model data unpublished
	primary erythermalgia SCN9A blockers (3)	**2016** Patient iPSC-derived sensory neurons (IEM) Not measured in patients	**2016** PBPK model from a human IV/oral microdose study (PF-05089771)	**2011** SCN9A human genetics (loss-of-function CIP, gain-of-function IEM)	**2010** Non-clinical toxicology (PF-05089771, funapide) Model data unpublished
Oncology	**B-cell non-Hodgkin lymphoma** MS4A1-directed depleters (3)	**2024** MB-CART20.1 product co-culture with CD20+ target cells No checkable prediction	**2005** Disseminated Daudi lymphoma in SCID mice No checkable prediction	**2020** Disseminated Daudi lymphoma in SCID mice	**2005** Disseminated Daudi lymphoma in SCID mice No checkable prediction
	Waldenstrom macroglobulinemia MS4A1-directed depleters (3)	**2009** Cynomolgus monkeys dosed with ofatumumab (B-cell depletion)	**2012** SCID mice bearing CD20+ human B-cell tumour xenografts dosed with ofatumumab (tumour…	**2012** SCID mice bearing CD20+ human B-cell tumour xenografts dosed with ofatumumab (tumour…	**2009** Cynomolgus monkey toxicology
	acute lymphoblastic leukemia CD19-directed depleters (6)	**2013** 19-28z CAR T cells in SCID-Beige mice with systemic NALM-6 ALL	**2013** Patient-derived ALL xenografts in NOD/SCID mice… No checkable prediction	**2021** Syngeneic murine anti-mouse-CD19 CAR T cells in… △ ORR 25.5%, trial stopped	**2021** Syngeneic murine anti-mouse-CD19 CAR T cells in lymphodepleted lymphoma-bearin…
	chronic lymphocytic… BTK inhibitors (4)	**2019** Mouse oral-dose BTK PD (occupancy probe, CD69/CD86)	**2022** Sponsor PBPK model (Simcyp) for CYP3A drug-interaction dosing	**2020** Spontaneous canine B-cell lymphoma (pet dogs)	**2022** Kinase selectivity panel and ex-vivo platelet/NK assays vs ibrutinib
	chronic lymphocytic… CD19-directed depleters (4)	**2014** Patient CLL cells with NK effectors (ex-vivo ADCC) △ Ibrutinib did not raise CAR-T…	**2020** Cynomolgus monkey repeat-dose toxicology (tafasitamab)	**2019** CD19 CAR-T in immunodeficient-mouse CD19+… △ No ORR gain over rituximab	**2020** Cynomolgus monkey repeat-dose toxicology (tafasitamab)
	chronic lymphocytic… MS4A1-directed depleters (4)	**2024** Cynomolgus monkey (B-cell depletion, cytokines, SC vs IV)	**2020** Cynomolgus monkey (B-cell depletion, cytokines, SC vs IV) No checkable prediction	**2021** Primary CLL cells from patients, ex-vivo ADCC/CDC vs rituximab	**2021** Cynomolgus monkey (B-cell depletion, cytokines, SC vs IV)
	chronic myeloid leukemia ABL1 inhibitors (3)	**2012** BCR-ABL cell lines and mouse leukemia/xenograft models	**2012** Ba/F3 BCR-ABL cell-based mutagenesis screen	**2020** BCR-ABL cell lines and mouse leukemia/xenograft models	**2025** Biochemical kinase-selectivity profiling (VEGFR2 and other… △ Heart, liver, pancreas flagged;…
	diffuse large B-cell lymphoma CD19-directed depleters (4)	**2017** Primary pre-B ALL xenograft in NSG mice (CTL019 CAR)	**2019** Cynomolgus monkey repeat-dose toxicology… No checkable prediction	**2021** Human lymphoma xenografts in immunodeficient mice (ADCT-402) △ No better than rituximab	**2021** Cynomolgus monkey repeat-dose toxicology (loncastuximab tesirine)
	diffuse large B-cell lymphoma MS4A1-directed depleters (5)	**2021** Cynomolgus monkey (epcoritamab toxicology/PD)	**2021** Mechanistic QSP model of CD20xCD3 CRS (mosunetuzumab)	**2026** Humanized-mouse lymphoma xenografts (CDX and DLBCL PDX) ☐ Outcome trial negative	**2021** Cynomolgus monkey (epcoritamab toxicology/PD)
	extranodal nasal NK/T cell… anti-PDCD1 antibodies (3)	☐ no model relied on	☐ no model relied on	**2020** EL4 murine T-cell lymphoma, syngeneic C57BL/6 mice, surrogate anti-mouse PD-1 (RMP1-14)	**2016** Cynomolgus monkey repeat-dose toxicology (tislelizumab) No checkable prediction
	gastrointestinal stromal tumor HSP90AA1/HSP90AB1 inhibitors (3)	**2014** GIST-PSW and GIST430 xenografts in SCID mice (onalespib ± imatinib)	**2012** BIIB021 xenograft PK/efficacy (AUC for ≥90% antitumour activity)	**2016** Imatinib-sensitive and… Ganetespib: no responses in refractory GIST	**2016** GIST-PSW and GIST430 xenografts in SCID mice (onalespib ± imatinib)
	large B-cell lymphoma CD19-directed depleters (3)	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells No checkable prediction	**2017** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells
	mantle cell lymphoma BTK inhibitors (4)	**2012** BTK active-site probe occupancy in mice, rats and dogs	**2012** BTK active-site probe occupancy in mice, rats and dogs	**2013** Pet dogs with spontaneous B-cell non-Hodgkin lymphoma	**2019** Zanubrutinib GLP toxicolog… Bleeding flagged, atrial fibrillation missed
	mantle cell lymphoma CD19-directed depleters (3)	**2020** Syngeneic mouse lymphoma, murine anti-mCD19 CAR T cells	**2021** Rat and cynomolgus monkey toxicology (loncastuximab tesirine)	**2020** Syngeneic mouse lymphoma, murine anti-mCD19 CAR T cells	**2020** Syngeneic mouse lymphoma, murine anti-mCD19 CAR T cells
	marginal zone lymphoma MS4A1-directed depleters (3)	**2025** IgM CD20xCD3 (imvotamab/IGM-2323 and cyno surrogate) in cynomolgus monkey…	**2026** Semi-mechanistic PBPK/PD model of epcoritamab trimer formation and tumour response…	**2017** Ex-vivo ADCC on patient leukaemia/lymphoma cells (ublituximab vs rituximab)	**2022** IgM CD20xCD3 (imvotamab/IGM-2323 and cyno surrogate) in cynomolgus monkey…
	myelodysplastic syndrome HDAC (pan) inhibitors (3)	**2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML)	**2011** Allometry and Simcyp PBPK (pracinostat) No checkable prediction	**2017** Mouse AML xenografts (MOLM-13, SET-2) CR 18% vs 33% placebo	**2015** Dog telemetry and rat/dog repeat-dose toxicology (panobinostat IND package)
	myelofibrosis JAK2 inhibitors (3)	**2019** Patient-derived JAK2V617F+ progenitor colonies and xenograft (TG101348)	**2008** JAK2V617F retroviral bone-marrow transplant mouse No checkable prediction	**2015** Patient-derived JAK2V617F+ progenitor colonies and xenograft (TG101348)	**2019** Nonclinical toxicology of… Wernicke's encephalopathy missed by tox
	small cell lung carcinoma anti-PDCD1 antibodies (3)	**2023** In-vitro PD-1 occupancy of toripalimab on human T cells, saturating cynomolgus tox	**2016** Cynomolgus tox, nivolumab	**2021** Tumour-mutational-burden rule, SCLC exomes, nivolumab △ Missed OS, indication withdrawn	**2017** Mycobacterium tuberculosis aerosol-infected PD-1 knockout… △ Fatal pneumonitis, not in monkeys
Ophthalmology	**diabetic macular edema** anti-VEGFA antibodies (3)	**2011** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal VEGF Trap, VGT-NC-007)	**2023** Rabbit chronic retinal-neovascularisation leak model (aflibercept 8 mg)	**2021** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal KH902/conbercept)	**2023** Monkey ocular tox, aflibercept PHOTON: no nasomucosal signal at 8 mg
	dry eye syndrome anti-TNF antibodies (3)	**2023** Aged mouse dry eye Not measured in patients	**2013** Rabbit topical ocular PK (ESBA105 eye drops)	**2022** Canine dry eye (spontaneous) Dog sign benefit; Phase 3 flat vs placebo	**2022** Rabbit topical ocular PK (systemic exposure margin)
	retinitis pigmentosa RPGR replacement / gene therapy (3)	**2024** Isogenic RPGR-knockout human iPSC retinal organoids,… No checkable prediction	**2024** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova)	**2026** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova) △ LUMEOS missed VMA primary	**2024** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova) △ Inflammation at higher doses,…
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold)	**2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2019** Mouse emotion-induced cataplexy (lemborexant) △ Animals no abuse; humans liked it
Respiratory	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants △ Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2025** Rat and dog tox (vanzacaftor) Patient result unclear
	cystic fibrosis CFTR potentiators (5)	**2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping)	**2022** Human hepatic metabolism in vitro, scaled to PK	**2024** FRT cells carrying each CFTR variant (Ussing chamber)	**2024** Rat toxicology (juvenile and adult, ivacaftor program)
	pulmonary arterial hypertension PTGIR agonists (3)	**1993** Human platelet-rich plasma in vitro, ADP-induced aggregation… △ No haemodynamic change in…	**2016** Rat MCT and Su5416/hypoxia, intratracheal… Model data unpublished	**2019** PAH-patient PASMC proliferation and human distal… BEAT: no effect on clinical worsening	**1994** Monocrotaline rat (oral dosing)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis Not measured in patients	**1997** Rodent chemical cystitis No checkable prediction	**2022** ASP6294 vs anti-NGF IgG in arthritic rodents ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox Joint harm missed; IC trials put on hold
Other	**atopic eczema** HRH4 antagonists (3)	**2014** Mouse FITC dermatitis, H4R	**2010** Mouse FITC dermatitis, H4R No checkable prediction	**2021** Dog acute atopic dermatitis, JNJ 7777120 and JNJ 28307474 △ Adriforant: mice yes, patients no	**2014** Rat, monkey 6/9-month tox Agranulocytosis the tox species never showed